AI/ML Problem Statement

Campus Problem Solver - Problem Statement

Team Size: 1-3 members

Submission Format: GitHub + Live Demo

🎯 THE CHALLENGE

Build an **AI Agent-powered campus problem resolver** that automatically classifies student complaints and routes them to the right department.

In simple terms:

- Student submits: "Bathroom on 3rd floor has no water"
- **AI Agent** classifies it as: **Bathroom Issue** → Routes to **Maintenance**
- Maintenance gets notified → Updates status
- Student sees live updates on their dashboard

📋 WHAT YOU NEED TO BUILD (MVP ONLY)

1. PROBLEM SUBMISSION PORTAL (Frontend)

→ Streamlit frontend also works

```
[Simple Web Form]
├─ Problem Description (text area)
├─ Upload Image (optional)
├─ Submit Button
└─ Tracking ID shown after submission
```

Must-Have:

- Form validation (empty field check)
- Displays submission confirmation with tracking ID
- Shows "Status: Submitted" on dashboard

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2. ML CLASSIFICATION AGENT (Backend + ML/AI)

- * You can use LLM also for this step...

```
[Student Input] → [NLP Preprocessing] → [AI Agent] → [Categ
ory Output]
```

Agent must:

- Classify into 6 categories minimum:
  1. Bathroom & Hygiene
  2. Anti-Ragging & Safety
  3. Mess & Food Quality
  4. Academic Issues
  5. Infrastructure/Maintenance
  6. Other
- Accuracy target: ≥75%
- Return classification + confidence score
- Simple fallback for low-confidence results

Note: Train agent on minimum 500 labeled problems (or use synthetic data) - This is not mandatory.

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3. AUTOMATIC ROUTING SYSTEM (Backend)

```
Classified Problem → Agent Routes → Auto-assign to Departme
nt → Send Notification
```

Must-Have:

- Database mapping: Category → Department/Executive
- Store problem with classification in database
- Notify assigned executive (email or log to admin panel)
- Simple routing logic (no complex rules needed)

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4. ADMIN/EXECUTIVE DASHBOARD (Frontend)

```
[Assigned Problems List]
├─ Problem description
├─ Category (auto-filled)
├─ Status (Submitted/In Progress/Resolved)
├─ Update Status Button
└─ Add Resolution Text
```

Must-Have:

- View all assigned problems
- Update problem status
- Add response/resolution comment
- Simple table or card layout

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5. USER TRACKING DASHBOARD (Frontend)

```
[My Problems]
├─ Problem #1 (Status: In Progress - Updated 2 hrs ago)
├─ Problem #2 (Status: Resolved - Dec 10)
└─ Problem #3 (Status: Submitted - Just now)
```

Must-Have:

- List submitted problems
- Show current status
- Show updates/responses from admin
- Real-time or refresh updates

✅ SUBMISSION REQUIREMENTS

- [ ] GitHub repository with complete code
- [ ] Live demo (Streamlit Cloud, Vercel, or local)
- [ ] README with setup & usage instructions
- [ ] Agent classification accuracy documented

- [ ] All 5 components working end-to-end

Good luck!